

Bodystorming for VR in the Dark: Using Glow Sticks for Ideation and Rapid Prototyping

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ABSTRACT

Exploration and rapid prototyping of design ideas for VR experiences is challenging for both novice and experienced designers as well as people that are not designers, especially in co-design projects. In this speculative paper, I propose *Glowing Bodies*, a bodystorming technique that leverages on the intrinsic immersive quality of a dark environment where participants use a simple prop, glow sticks, to create an embodied ideation and rapid prototyping space for VR.

Keywords: Virtual reality, immersive technologies, design ideation, participatory design, XR experiences.

Index Terms: •Human-centered computing~Interaction design~Interaction design process and methods

1 INTRODUCTION

Consumer level headsets that offer virtual reality (VR) experiences are considered as an opportunity to introduce innovative changes or novel ways of doing things in various fields such as entertainment, education, medicine, architecture and so on. However, exploration and rapid prototyping of design ideas for VR experiences is still challenging for both novice and experienced designers as well as people that are not designers.

One of the reasons for this is the landscape of immersive technologies and how it rapidly changes. In terms of technical setup, hardware and software present some obstacles such as clunky headsets, cables restricting bodily movement, complicated application interfaces, operating costs that prevent design teams from doing quick ideation or rapid prototyping, in an individual or collective setting, to explore design challenges in immersive spaces. Regarding the pace of change in technology, some applications offering a potential platform for sketching and prototyping immersive ideas might be discontinued or become obsolete after a relatively short time [6,13]. In addition, the threshold for learning new software to sketch and prototype design ideas for virtual environments might be high even for designers. This is especially a challenge in co-design projects where teams comprise individuals with different technical skills and knowledge such as stakeholders that are not familiar with digital technologies or immersive experiences.

Another reason is that traditional ideation and sketching techniques such as handdrawn or digital storyboards do not allow for exploring the spatial and bodily aspects of VR experiences in an embodied way nor for capturing its interactive potential. Whereas VR requires designers and stakeholders to sensitize themselves to the virtual space—the feeling of being there—for which they develop a design concept by building environments, objects, stories, characters, and interactions. [2] hints at this bodily and physical dimension, reporting that roleplaying is considered more effective for prototyping XR experiences, as it enables participants—also those who are not familiar with the world of XR—to illustrate their thoughts in an easier and faster way.

So, I argue that there is a need for low-threshold sketching and rapid prototyping techniques for VR which are embodied and light-weight; techniques that would be useful especially in early stages of co-design projects where a diverse group of participants work together to sensitize themselves to immersive spaces and explore design ideas. In this speculative paper, I propose *Glowing Bodies*, a bodystorming technique that is based on using glow sticks as building blocks and leverages on the intrinsic immersive quality of a dark environment. What this means is that we are not able to see our surroundings in the dark, which forces us to guess where things might be. Not being able to see spatial cues or objects from the real-world, and only things built with glow sticks, we start taking them as reference points to orient ourselves in space. This results in a blank three-dimensional canvas that creates the illusion of an immersive space where participants are encouraged to use their imagination and a simple prop, glow sticks, to build VR objects and environments. In this way, glow sticks transform the dark into an embodied ideation and rapid prototyping space for VR. More specifically, participants are able to do two things: (a) sketching and prototyping their ideas by using glow sticks to build characters, objects and environments in real-time; (b) visualizing and documenting possible VR interactions and their visual representations through light paintings created by bodily movements or mobile photography. I will first describe the technique in detail and then discuss why it might be meaningful to adopt it in a co-design process, referring to its advantages as well as limitations.

2 RELATED WORK

Bodystorming is an established method in interaction design, which is used to gain a more personal, deeper understanding of a design situation by being in the context of interest to some degree and scale [3,4,11,12]. There are various interpretations and uses of bodystorming. The basic idea behind the method is to be in the actual location, or a representative one, to study the context of interest through embodied action in space instead of sitting in an office or lab space reading documents and reasoning about possible challenges. In this paper, I adopt the bodystorming approach as described by Schleicher et al. [12] because of its explorative nature; the authors highlight the importance of joint acting and improvisation through scenarios to create a rapid cycle of communication, which aims to result in rapidly simulating experiences and generating ideas. These authors also argue that such bodystorming would be beneficial in the problem-definition stage. I would like to highlight that the paper does not propose *Glowing Bodies* as a new method that substitutes bodystorming but rather as a technique that could be used in bodystorming for VR. In this sense, *Glowing Bodies* can be seen as “a particular form of expression and execution”, taking the description of what a technique is in [9].

There are tools and techniques that allow for live sketching in space for extended reality (XR) [1,8]. However, these are based on using devices such as a VR/MR headset, tablet, or stylus to do sketching. This requires access to specific equipment and learning how to sketch with them, which does not make them suitable for rapid prototyping. One prototyping technique is based

on making 2D paper mockups of XR assets and spaces that can be transformed into the intended environment, which enables previewing, organizing sequences and adding interactivity, and testing the resulting experience [10]. However, this technique does not allow for embodied sketching and ideation, as XR elements are created through desk sketching. Moreover, it is tied to the use of specific software for immersive preview and detailing that increases the threshold in terms of accessibility and learning.

The prototyping technique for XR that I find closest to Glowing Bodies in the literature is Immersive Video Sketching (IVS), which combines bodystorming, paper prototyping and video sketching [5]. In this technique, participants first prepare paper prototypes and a draft scenario. Then, they enact the storyline with paper props while one person in the group takes 360-degree pictures of these enactments. Finally, they edit the pictures adding visuals representing virtual objects, actions, interface elements, speech bubbles etc and create interaction sequences in online 360-degree tour creation platforms. This technique introduces a workflow that enables participants to document their performances and create immersive sequences right after. In this way, they can experience their concepts in the intended 360-degree medium and communicate it to others. The difference between two techniques might be one of focus. IVS seems to put an emphasis on enacting scenarios with diegetic prototypes prepared ahead of the session and working towards documenting a concept. Glowing Bodies, instead, focuses on bodystorming supported by simple props (glow sticks) to prototype objects and environments according to situations that emerge through improvised exploration of the blank space. Adding new objects becomes part of diegetic action that unfolds a story, like in improv theater. In other words, in Glowing Bodies, the creation of an immersive environment happens in real-time as bodystorming unfolds in a space where participants have minimal visual connection to the real-world setting.

Similar to IVS, Glowing Bodies allows participants to document their process by taking panoramic or 360-degree pictures with smartphones or 360-degree cameras. However, one difference of Glowing Bodies is that it affords light painting because of the dark environment. Light painting is an effect created by long exposure photography; when a person moves a light source in the air, as if drawing a circle, its movements in space are captured by the camera which results in a picture where the circle made by the light source is visible. Being able to light paint makes it possible to create stills or immersive sequences in action as bodystorming happens, without the need for a deliberate post-production phase or a photo editing software.

3 DESCRIPTION OF THE TECHNIQUE

Glow sticks—also called light sticks—are objects that emit light when bended because of a resulting chemical reaction this action triggers. Once they are lit, they last for a few hours. These sticks are preferred as light sources at concerts, parties or military operations because they do not require electricity or batteries and work underwater. The technique is inspired by the videos in which people dance in the dark with glow sticks attached to their clothes. Because it is dark, and the sticks correspond to the limbs, one watching the video does not see anything else but a glowing stick figure [14]. The following text presents a fictional co-design workshop to provide a speculative account on what the technique is and how to apply it:

Let us imagine that the project participants—teamed up to work on designing a social reading space in VR—come together in a large design studio or meeting space. The team comprises an interaction designer (also moderator of the workshop), software engineer, librarian, library manager, and two avid readers of paper books. It is the second time the participants see each other, and the aim of the meeting is to make them familiar with immersive spaces and explore together what kind of virtual environment is envisioned for a social reading space. For example, what type of interactions are imagined between people visiting this space, what kind of interactions are imagined between visitors and the books, what visual form the books will have and what kind of reading experiences they will offer and so on.

The designer introduces Glowing Bodies, and assigns to one from the group the task of taking pictures during the session. The meeting room is prepared: there are no tables in the center of the room which gives enough space for the participants to move around. There are some soft objects lying around such as cardboard boxes, pillows, toys and so on. She shows a box of glowing sticks and asks the participants to use them to turn themselves into a glowing stick figure; maybe as the figure of an imagined visitor of the virtual reading space. After attaching the sticks on their clothes, they are ready for the exploration (Figure 1). The designer turns off the light, and the whole room gets pitch black. However, the participants are able to see themselves as future visitors. Not each other's real faces, but the faces they created bending the glow sticks in the form of eyes, nose, mouth etc. Some participants just move in this dark space, imagining possible spots or activities in the reading space; seeing them in the mind's eye. They also have the possibility of rapidly creating paths, spots, objects by placing the glow sticks on the floor or attaching them to the soft objects lying around. The librarian takes a few sticks to make an object resembling a book with a few lines. Then, one of the avid readers joins the librarian's sketching/prototyping process and grasps a few lines from the "book", makes them fly in the air for a while and land on another corner of the immersive space, suggesting a magical interaction between the book content and visitors. The participants sometimes talk to each other and sometimes remain silent, enjoying the tacit sharing of experiences in emerging situations. Sometimes the designer asks a question which might be relevant for the topic of a virtual reading space, or a question prepared earlier based on desk research or observations. Within the same session, the task of taking pictures can be handed over in rounds to another team member so that everyone has the chance to participate in the joint exploration and ideation.

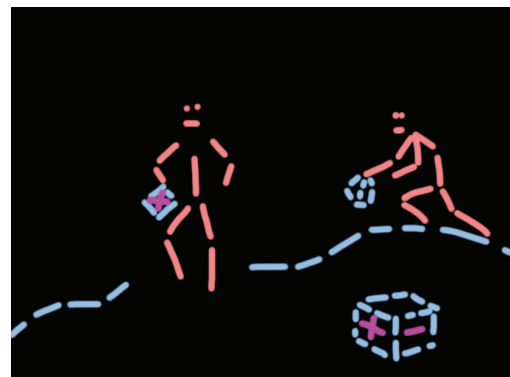


Figure 1: Sketch showing two participants that are exploring the space where they created a few objects with glow sticks.

The team needs to get a feeling of the virtual space they are designing for. Ideally, this should take place in a responsive environment where each team member has the possibility to see and explore how bodies should move in the virtual space and interact with each other and their surroundings in real-time. Glowing Bodies aims to do this. Based on a material (glow stick) that does not require technical skills to interact with allows participants to use their existing real-world skills to ideate as they move in this space and manipulate objects. Using glow sticks as light and color sources in a completely dark room they create the illusion of a virtual environment in the real world. In this way, they can experience how bodies can move, act, and interact with their surroundings as well as what kind of stories can be told in this immersive environment and other relevant questions that they identified earlier or that will emerge as they improvise.

After an hour or so, the bodystorming session ends. The designer turns the lights on. The whole session was recorded on video. The team watches the video recording in a retrospective think-aloud approach to reflect over and discuss their actions

during the bodystorming. After the think-aloud session, participants may look at the panoramic or spherical pictures they have taken and create immersive sequences (like in the IVS technique mentioned earlier), importing them into a free 360-degree media viewer (e.g., Marzipano.net) to have a quick impression of the environment this time in VR. This session aims to help them reflect in action over the topic or problem in question, gather insights about relevant design challenges, and possibly come up with some ideas on how to (re)define or make them more specific. After *Glowing Bodies*, they are hopefully more familiar with possible environments for a virtual reading space, and with what people visiting this place would feel and do in this environment. This session also enabled them to get closer to each other thanks to the playful atmosphere of the activity.

4 DISCUSSION

This technique enables us to create a space where people can simulate an artificially created environment in the real-world, and this space would afford exploring both traditional and magical interactions in VR thanks to the quick and easy re-arrangeability of glow sticks. Participants feel as if they are in a virtual reality space while they are in full control of their physical bodies. They are exploring the virtual in the physical space. The technique also creates an interesting situation: exploring the artificial/virtual world in a real-world environment thanks to the graphic quality of glow sticks. The material itself has a visual quality that resembles certain styles associated partly with computer graphics, which might be seen also in (for example, wireframes, vector graphics, or the visual style of the movie *Tron*). The visual character of this technique might not fit the visual style of all design projects. However, it would still enable people to explore spatial and social interactions, and how different visual components such as icons, menu items, feedback visuals etc should behave. *Glowing Bodies* might not be suitable to use in all types of XR; I think that it works best for working on VR environments. That said, the technique might be developed further to enable sketching for mixed reality experiences; for example, using a smartphone flashlight can provide a glimpse of the real-world when it is needed to see or contact the physical environment as part of a design concept.

Writing in the air with glow sticks can leave traces that can communicate new ideas or trigger thoughts by offering visual stimuli (Figure 2). These embodied writings can be done to answer a series of design questions prepared earlier or the ones that emerge as participants improvise.



Figure 2: Sketch showing participants that are doing light painting using glow sticks to sketch icons or feedback visuals.

Bodystorming for VR in the real-world would probably have its advantages, but it might be also highly cluttered, as there would be several objects around that are maybe not relevant to the design situation in hand. The fact that there is no visible clutter in the environment, and that the participants start from a blank page, so to speak, opens up space for imagination: ambiguity inherent to the dark environment suggests objects, interactions, movements

etc. Black light theater can be used as an inspiration and performance knowledge to inform the design of *Glowing Bodies* sessions and further develop the technique.

There would be a few advantages to this technique. For example, to have full body-representation in VR requires expensive and/or complicated tracking solutions, as people in their headsets do not see each other's physical bodies. In the glow stick-driven environment, they may see the bodies of participants in much higher resolution (both visual and temporal) when it comes to bodily movements without the need of specialized digital components. This might help create a more immersive experience than sketching in VR as the bodies are represented in kind of a higher interaction fidelity while still at an visual abstraction level enabling imagination. Starting from the dark and using glow sticks as building blocks for creating a VR environment with virtual objects offers a low-threshold activity for novice/non-tech-savvy participants. In addition, glow sticks provide abstract enough visuals to stimulate creativity and encourage interpretation and play, which might embrace both adults and children as participants of this activity. An interesting and potentially useful constraint of glow sticks is their limited glowing time (around two hours depending on the brand and type), which would determine how long a session will last and how it should be structured accordingly.

When it comes to the roles in the design team, while most participants can use glow sticks to turn themselves into stick figures, one person might act as a visual maker who, according to emerging situations, would create visual objects or feedback in real-time. Moreover, participants can also use their own bodies in a way that they enact or represent objects such as a window, door, table etc. The kind of experience *Glowing Bodies* offers has the potential to enable participants to build empathy towards characters in a VR experience to be designed. Although the technique may be more suitable for design teams or co-design settings, it can also be useful in one person exploration of VR spaces.

Regarding social dynamics, the darkness might help participants feel at ease, make them remain anonymous and thus perform in a more relaxed way. As people will not be seeing each other's faces clearly, they might get into their roles more quickly and easily, and also tend to express their ideas more openly. This may also help reduce power asymmetries that usually emerge when a group of diverse members ideate together. For example, if people do not see each other before turning the lights off, they would not understand who is who if they do not speak. Although not directly social, motion sickness is a factor that prevents people from using VR headsets. If people are motion sick while designing in VR they might find this method useful, as they are not required to wear a headset for participating in joint ideation.

When it comes to the techniques' relationship to digital technologies, one aspect might be of interest: bodily movements in *Glowing Bodies* sessions might be recorded or tracked live to process later: for example, using them as motion guidance for animated characters for purposes of presentation or further ideation or refinement.

Finally, reflecting over existing prototyping techniques for VR, one limitation of glow sticks is that they might not be as flexible and high-resolution as paper, but they might be more suggestive and can be used in combination with other objects (because of the darkness all objects become potential props, as they will not be recognized unless participants come close—and even then the perception of the object used as a prop might trigger ideas about an on-going question or design challenge in exploration. A material limitation and concern about the technique is that glow sticks do not seem to be sustainable. They cannot be reused, and they might not be recyclable. This downside might be partly overcome by upcycling them as materials for physical prototyping of new design ideas. On the other hand, there are also efforts to make glow sticks with eco-friendly materials [7].

5 CONCLUSION AND FUTURE WORK

In this speculative paper, I proposed a bodystorming technique, *Glowing Bodies*, that can be used for rapid sketching

and prototyping VR experiences. The technique leverages on the intrinsic immersive quality of dark environments and is based on using a simple prop, glow stick, to build objects and spaces in the real-world. It does not require to use headsets, drawing software and similar technological devices to experience a VR environment, but an item that participants can easily use without any prior technical knowledge. In this way, it presents a low-threshold technique that can be especially effective in co-design projects.

For future work, I plan to run a couple of bodystorming sessions using this technique in a course I am teaching on immersive information spaces. In the first step, our aim is to use this technique as an exercise to teach students to think and sketch through bodies and space for VR. Using this technique in a course will enable me to understand how it would work in a real-life setting, what participants think about it, and what is needed to structure it in line with intended effects in a design process. In the second step, the technique can be used in a real-life co-design project where a diverse group of people work on designing a VR experience. A session in such a setting would enable me to see how the technique plays out in accommodating different technological skills and knowledge, creating interaction between participants and space, and facilitating joint exploration of a topic or design challenge.

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